

Block 3 — Chapter 1: Safety, Ground Operations & Servicing

Plain-English Study Guide

amt-tutor make-up study session · 2026-06-15 · taught at 5th-grade-teacher level, nothing dumbed-out

Source (read in full, source of truth): FAA-H-8083-30B (Aviation Maintenance Technician Handbook—General), **Chapter 1 — Safety, Ground Operations, and Servicing**, printed pages 1-1 through 1-30 (your file “Chapter 1 Ground Ops.pdf”). Every fact below is cited to its printed page (e.g. “p. 1-14”).

How to use this: read this instead of the whole chapter. The yellow MUST KNOW boxes are the items most likely to show up on your class quiz and the cumulative block exam — memorize those cold.

PART 1 — Shop Safety

The whole chapter starts with one idea: a clean, orderly shop is a safe shop. Keep the shop, hangars, and flight line clean; the outgoing shift stows tools, stands, hoses, and cords. Signs mark dangerous equipment and where first-aid/fire gear is. Painted **safety lanes, walkways, and fire lanes** keep foot traffic out of work areas. [p. 1-2]

Electrical Safety

Electricity through the human body causes **burns at the point it enters and the point it exits**, and it can damage or destroy the **nervous system**. [p. 1-2] To work safely you need a working knowledge of electricity AND a healthy respect for it.

- **Two attitudes cause electrical accidents: fear and overconfidence.** A little respect is healthy; too much of either is deadly. Lack of respect usually comes from lack of knowledge. [p. 1-2]
- Safety gear that protects you (and gives peace of mind): **rubber gloves, safety glasses, rubber or grounded safety mats.** [p. 1-2]
- **Electrical fire prevention:** keep the area clean, uncluttered, and free of flammables. Keep cords free of kinks/bends. Never put cords where they get walked on or run over. If wires inside a cord break, the remaining wires carry more current → more heat → fire. [p. 1-2]

5th-grade picture: A wire is like a water hose. If half the strands inside break, the leftover strands have to carry all the “water” (current) — they get overworked and hot, just like pinching a hose makes pressure spike. Too hot = the insulation melts and starts a fire.

Safety Around Compressed Gases

Compressed air, like electricity, is a great tool only when it is under control. [p. 1-2] The dos and don'ts:

- Inspect air hoses often for breaks/worn spots; replace bad hoses immediately. Keep connections in a “no-leak” condition. [p. 1-2]
- **Never use compressed air to clean hands or clothing** — pressure forces debris into the flesh and causes infection. Never spray it toward other people. [p. 1-2]
- Coil and store hoses when not in use. Filter air used for paint spraying to remove oil and water. [p. 1-2]
- **Tire mounting / inflation:** many compressed-gas accidents happen mounting tires. Use tire dollies for heavy tires. When inflating, **always use tire cage guards** and a **pressure regulator** on high-pressure bottles to avoid over-inflation. **Tire cages are NOT required when adjusting pressure in tires still installed on an aircraft.** [p. 1-3]



Figure 1-1. A typical nitrogen bottle.

Safety Around Hazardous Materials — where the hazard info lives

Two things tell you a chemical's danger:

1. **The risk diamond (material safety diamond)** — a four-color segmented diamond: **red = flammability**, **yellow = reactivity**, **blue = health**, **white = special hazard**. The red/yellow/blue blocks hold a number **0 to 4** (0 = little/no hazard, **4 = very hazardous**). The white block holds a word/abbreviation: RAD = radiation, ALK = alkali, Acid = acidic, CARC = carcinogenic, and a **W with a line through it = reacts strongly with water**. [p. 1-3]
2. **The SDS (Safety Data Sheet)** — the detailed version: full chemical breakdown, formulas, and what to do if you contact the chemical. **Formerly called the MSDS** (Material Safety Data Sheet). The required information is the same on every sheet, but its exact location can vary by

manufacturer. SDSs are required by the governing safety body, **OSHA** (the U.S. Department of Labor’s Occupational Safety and Health Administration). [p. 1-3]



Figure 1-2. A risk diamond (NFPA 704).

MUST KNOW **Where to find hazard info:** quick look = the **risk diamond** (red flammability / yellow reactivity / blue health / white special, numbered **0–4**, 4 = worst). Detailed info = the **SDS** (Safety Data Sheet), **formerly the MSDS**, required by **OSHA**. [p. 1-3]

Safety Around Machine Tools

Each machine has its own rules, but they share a backbone: **wear eye protection, secure the work, set the correct RPM, and clean up after.** [pp. 1-3 to 1-4]

Machine	What it does	Key safety points
Drill press	Bore/ream holes, facing, milling	Eye protection; clamp work securely; set proper RPM; don't feed spindle past its travel limit; stop the machine before adjusting or clearing jammed work.
Lathe	Turns cylindrical work (inside or outside)	Eye protection; sharp tools; let the chuck stop on its own — never stop it by hand; don't set tools on the lathe; let work stop before measuring.
Milling machine	Shape/dress, cut gear teeth, slots, keyways	Eye protection; clean and secure the work bed; don't change feed speed while working; lower the table before moving away.
Grinder	Sharpen tools, dress metal, remove small amounts	Eye protection even with a shield; inspect the wheel for defects first; never force a wheel onto the spindle (it can explode); flanges/washer = 1/3 the wheel diameter; don't stand in the wheel's arc in case it explodes.

Welding (a special shop hazard)

Weld only in **designated areas**; remove the part from the aircraft if possible and weld in a controlled shop. If you must weld in the hangar: no open fuel tanks / fuel-system work, no painting, **no aircraft within 35 feet**, no flammable materials nearby, only qualified welders, rope off and placard the area, keep **fire extinguishing gear rated at least 20B in the immediate area with 80B as backup**, a trained fire watch present, and the aircraft kept towable with a tug attached and brakes released. [p. 1-4]

PART 2 — Flight-Line Safety

Hearing Protection — why and when

The flight line is loud from many sources at once — aircraft, **APUs, fuel trucks, baggage equipment** — and combined that noise can cause hearing loss. [pp. 1-4 to 1-5]

- **External** protection = earmuffs / headphones. **Internal** protection = plugs that fit into the auditory (ear) canal. Both lower the sound reaching the eardrum. [p. 1-5]
- **When it is essential:** working with **pneumatic drills, rivet guns, or other loud tools**. Even **short exposure** to these high-frequency sounds can cause hearing loss; continued exposure *will*. [p. 1-5]

Foreign Object Damage (FOD) — what it is and how to prevent it

FOD is any damage to aircraft, personnel, or equipment caused by any loose object — broken runway concrete, shop towels, safety wire, etc. [p. 1-5]

- **Prevention:** keep ramp/operation areas clean, run a **tool control program**, and provide convenient receptacles for used hardware, shop towels, and consumables. [p. 1-5]
- **The turbine-engine reason it matters:** a running gas turbine creates a **low-pressure area in front of the engine that sucks loose objects in**; the exhaust can throw objects far enough to damage anything they hit. Never leave tools or items near a turbine intake. [p. 1-5]

MUST KNOW **FOD = Foreign Object Damage** = damage to aircraft, people, or equipment from **any loose object**. Prevent it with a clean ramp, a **tool control program**, and receptacles for trash/hardware. [p. 1-5]

Safety Around Airplanes (the danger zones)

- **Propellers:** never assume a taxiing pilot can see you. **Stay within the pilot's view** on the ramp. [p. 1-5]
- **Turbine intakes (suck zone) and exhaust (blast zone) are very hazardous.** [p. 1-5]

- No smoking / open flames anywhere near an operating aircraft. Watch for aircraft fluids that harm skin. Keep ground equipment spaced from the aircraft and secured so it can't roll into it; stow loose items. [p. 1-5]

The chapter maps the danger zones around a turbine aircraft — the **suck zone** in front of each inlet and the **blast zone** behind. Notice the **25-foot** danger markers at the intakes.

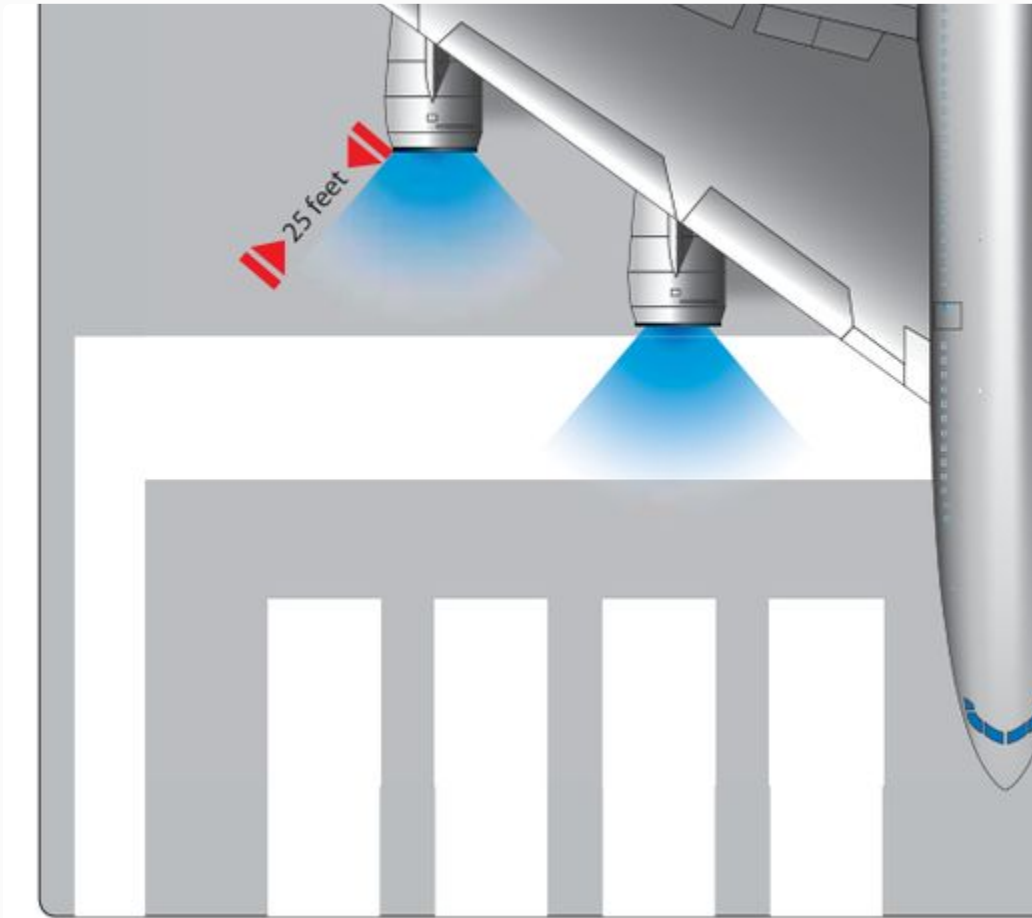


Figure 1-17. Engine intake and exhaust hazard areas.

Safety Around Helicopters — rotor approach rules

Every helicopter is different, so learn the type. When approaching while the blades turn: [p. 1-5]

1. **Check that the rotor head and blades are level** — that gives maximum clearance as you approach.
2. **Approach in view of the pilot.**
3. **Never carry anything tall (vertical height)** that the blades could hit — that damages the blade and injures you.
4. **Never approach a single-rotor helicopter from the rear** — the spinning **tail rotor is invisible**.

5. **Never cross from one side to the other by going around the tail. Always go around the NOSE.**

MUST KNOW Helicopter golden rules: approach **in the pilot's view**, when the rotor is level; **NEVER approach from the rear** (invisible tail rotor); to change sides, **go around the NOSE, never the tail.** [p. 1-5]

Fire Safety on the flight line

Maintenance uses spark-making tools, heat, and flammable liquids/gases, so the fire potential is high. **The key to fire safety is knowing what causes a fire, how to prevent it, and how to put it out.** [p. 1-5] That leads straight into Fire Protection.

PART 3 — Fire Protection

What a fire needs (the fire triangle)

Three things are required for a fire. **Remove any one and the fire goes out:** [p. 1-5]

1. **Fuel** — combines with oxygen in the presence of heat, releasing more heat.
2. **Heat** — speeds the combining of oxygen with fuel, releasing more heat.
3. **Oxygen** — the element that combines chemically with the fuel (oxidation). Rapid oxidation with noticeable heat and light = **combustion / burning.**

5th-grade picture: Think of a three-legged stool. Fuel, heat, and oxygen are the three legs. Knock out ANY one leg and the stool (the fire) falls over. That is literally how every extinguisher works — it removes one leg.

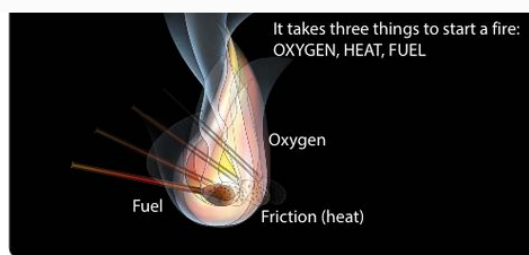


Figure 1-3. The three elements of fire (fire triangle).

Note on the “tetrahedron”: some texts add a 4th item, the chemical chain reaction, to make a fire *tetrahedron* (4 sides). This FAA chapter teaches the **three-element triangle** (oxygen, heat, fuel) — that is what it pictures and what your worksheet follows. [p. 1-5, Fig. 1-3]

The classes of fire — A, B, C, D

The National Fire Protection Association (NFPA) names three basic classes (A, B, C); a fourth (D) covers burning metal. [p. 1-6]

Class	What burns	Best extinguisher
Class A	Ordinary combustibles — wood, cloth, paper, upholstery	Water (best); also CO ₂ and multipurpose dry chemical
Class B	Flammable petroleum products / liquids — gas, oil, grease, solvents, paints	CO₂, Halon (1211/1301), dry chemical, foam — NOT plain water (fuel floats on water)
Class C	Energized electrical wiring and equipment	CO₂ or Halon (non-conductive) — water only with extreme caution after power is OFF
Class D	Flammable metal — magnesium (shop, wheels/brakes), bad welding	Dry powder ONLY — NEVER water or CO ₂

MUST KNOW **Fire classes (memorize letter → what burns):** **A** = ordinary combustibles (wood/paper/cloth) · **B** = flammable liquids (fuel/oil/grease/paint) · **C** = energized electrical · **D** = flammable metal (magnesium). Class D is “not commercially a basic type” per NFPA because it’s caused by an A/B/C fire. [p. 1-6]

Extinguisher types — how each works and what to use it on

- **Water** — best on **Class A**. Two effects: deprives the fire of oxygen AND cools the burning material. **Not for Class B** (petroleum floats on water). Use on Class C only with *extreme* caution — all power off first, and beware residual charge in capacitors/coils. [p. 1-6]
- **Carbon dioxide (CO₂)** — for **Class A, B, and C**. Smothers the fire (depletes oxygen) and cools it. Self-expelling (it has enough pressure to push itself out). Cautions: parts get **extremely cold** — **frostbite risk**; in confined spaces it can suffocate the operator too. [pp. 1-6 to 1-7]
- **Halogenated hydrocarbon (Halon, industry “Freon”)** — most effective on **Class B and C** (can be used on A and D but less effective). Works by chemically interfering with combustion and blanketing out oxygen, and resists re-flash. [p. 1-7]
- **Dry powder** — effective on B and C but **best for Class D (metal) fires**. **Not recommended for general aircraft use** (except metal fires) because leftover residue/dust is hard to clean and damages delicate equipment. [pp. 1-7 to 1-8]



Figure 1-4. Carbon dioxide (CO₂) fire extinguisher.

	A	
Water and antifreeze	X	
Soda-acid (water)	X	
Wetting agent	X	
Foam	X	
Loaded stream	X	
Multipurpose dry chemical	X+	
Carbon dioxide		
Dry chemical		
Bromotrifluoromethane — Halon 1301		
Bromochlorodifluoromethane — Halon 1211		
Dry powder (metal fires)		
+ Smaller sizes of these extinguishers are not recognized for use		

Figure 1-5. Extinguisher operation and methods of expelling.

Exam-trap callouts (the wording the FAA loves):

- **NEVER use water OR CO₂ on a Class D (metal) fire.** Their cooling effect makes the hot metal expand explosively — metals burn at extremely high temperatures. [p. 1-6]
- **Carbon tetrachloride (Halon 104) is NOT approved for any fire-extinguishing use.** It is extremely toxic and, near hot metal, produces **phosgene gas** (plus hydrochloric acid vapor and chlorine). [p. 1-8]

- **Halon 1301** = Bromotrifluoromethane (vapor spray, harder to direct, least toxic, UL rating 6). **Halon 1211** = Bromochlorodifluoromethane (forms a spray like CO₂, UL rating 5). The EPA restricted Halon to its 1986 production level (ozone). [pp. 1-7 to 1-8]
- **Toxicity table:** a HIGHER UL/group number = LESS toxic (6 = least toxic). Don't read it backwards. [p. 1-7, Fig. 1-6]

6 (Least toxic)	Gases or vapors in concentrations up to 20% by volume, for durations of exposure of up to approximately 2 hours, do not appear to produce injury.	Bromotrifluoromethane (Halon 1301)
5a	Gases or vapors much less toxic than Group 4, but more toxic than Group 6.	Carbon dioxide
4	Gases or vapors in concentrations of the order of 2 to 2 ½%, for durations of exposure of up to approximately 2 hours are lethal or produce serious injury.	Dibromodifluoromethane (Halon 1202)
3	Gases or vapors in concentrations of the order of 2 to 2 ½%, for durations of exposure of the order of 1 hour are lethal or produce serious injury.	Bromochloromethane (Halon 1011) Carbon tetrachloride (Halon 104)
2	Gases or vapors in concentrations of approximately ½ to 1%, for durations of exposure of up to approximately ½ hour are lethal or produce serious injury.	Methyl bromide (Halon 1001)

Figure 1-6. Toxicity table (higher group number = less toxic).

Inspection of fire extinguishers

Check extinguishers periodically with a checklist. If no checklist, at minimum verify: [p. 1-8]

- Proper location of the appropriate extinguisher · safety seals unbroken · all external dirt/rust removed
- Gauge/indicator in the operable range · proper weight · no nozzle obstruction · no obvious damage

Identifying an extinguisher (the markings)

Extinguishers are marked to show which fire class they suit, and the same marking goes in a conspicuous spot near the extinguisher. A location marked for Class B must actually hold a Class-B-suitable extinguisher. Markings (decals, paint) must be legible at the required distance — on the extinguisher itself, readable at **at least 3 feet**; on a wall panel, readable at **at least 25 feet**.

[pp. 1-8 to 1-9]

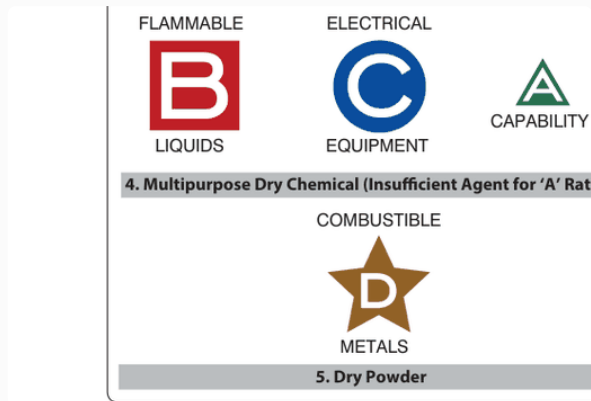


Figure 1-7. Typical fire-extinguisher markings.



Figure 1-8. Identification of fire-extinguisher type / location.

How to USE an extinguisher — the PASS method

The chapter spells out the steps; the standard memory word is **PASS**:

Letter	Step	The chapter's wording
P	Pull the pin	“Most extinguishers have a pin to pull that allows the handle to activate the agent.”
A	Aim at the base of the fire	“Stand back 8 feet and aim at the base of the fire or flames.”
S	Squeeze the lever	“Squeeze the lever...”
S	Sweep side to side	“...and sweep side to side until the fire is extinguished.”

MUST KNOW **PASS** = **P**ull the pin · **A**im at the **base** of the fire (from about **8 feet** back) · **S**queeze the lever · **S**weep side to side. And before all that: confirm it's the **correct type for that class of fire**. [p. 1-9]

PART 4 — Tie-Down Procedures

Preparing the aircraft: tie aircraft down after each flight to prevent storm damage. Park/tie-down direction is set by the prevailing or forecast wind — **head the aircraft into the wind**, allowing ample wingtip clearance between tie-downs. After it's positioned, **lock the nosewheel or tailwheel in the fore-and-aft position**. [pp. 1-9 to 1-10]

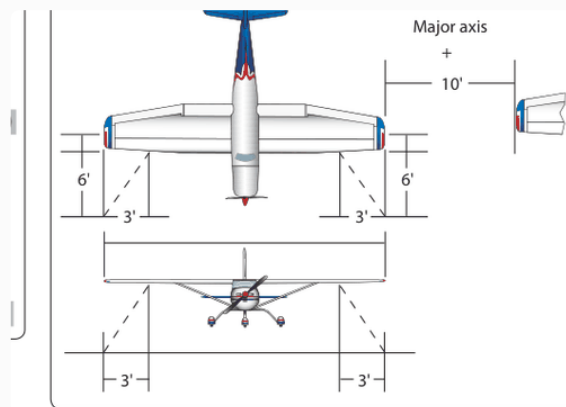


Figure 1-9. Diagram of tie-down dimensions.

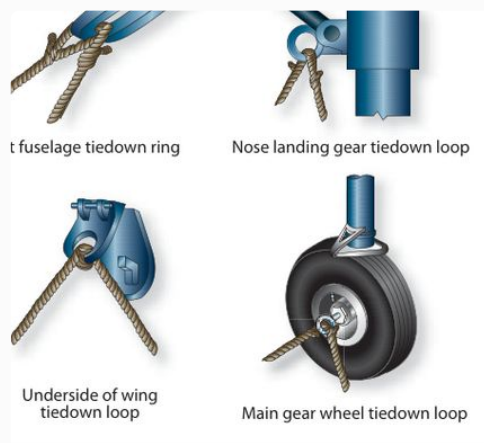


Figure 1-11. Common tie-down points.

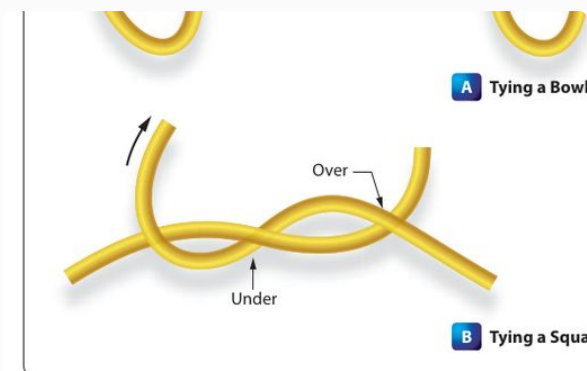


Figure 1-10. Knots commonly used for aircraft tie-down.



Figure 1-12. Wheels chocked fore and aft.

Aircraft type	How to secure it
Land plane (light)	Tie ropes only to the provided tie-down rings — never to a lift strut (a slipping rope can bend the strut). Manila rope shrinks when wet, so leave about 1 inch of slack — too much slack lets it jerk; too tight puts inverted-flight stress on the airframe. Use an anti-slip knot like a bowline . [p. 1-10]
Land plane (heavy)	Rope or cable; number of tie-downs depends on weather. Most have surface control locks — install them; add control-surface battens for high wind. Procedure: head into wind → install control locks/covers/guards → chock all wheels fore and aft → attach tie-down reels (or 1/4" wire cable / 1-1/2" manila line). Use tie-down stakes for temporary only. [p. 1-11]
Seaplane	Moor to a buoy or tie to a dock (weather permitting); waves make it bob/roll, which can cause damage. In a storm you can't fly out of, flood the floats with water and tie down; if possible, remove from the water and tie like a land plane, or anchor in a sheltered area. [pp. 1-10 to 1-11]
Ski plane	Tie down like a land plane if anchors exist. On ice/snow, use a " dead-man " — any item (pipe, log) tied to a rope and buried in a packed snow/ice trench (pour water and

Aircraft type	How to secure it
	let it freeze). You can also freeze the skis to the ice. Caution: make sure a ski isn't still frozen down before moving the aircraft. [p. 1-11]
Helicopter	Hangar it if possible; tied down it survives winds up to ~65 mph . Evacuate for tornadoes/hurricanes. Mooring: (1) face it into the highest forecast wind/gusts; (2) spot it slightly more than one rotor span from other aircraft; (3) chock all wheels (or lower onto skids and pin); (4) align blades and install tie-down assemblies per the manufacturer (snug, with slack in wet weather); (5) fasten ropes/cables to the forward and aft landing-gear cross tubes. Tie down the main rotor blades if high winds are expected. [pp. 1-11 to 1-12]
Weight-shift-control	Very wind-susceptible. In light wind, secure the wing like a conventional aircraft. In high wind, disconnect the mast from the wing, lay the wing near the ground and secure it , or partially disassemble / hangar it. [p. 1-12]
Powered parachute	Pack the parachute (chute) in a bag so wind can't fill it and drag the seat and engine. Secure the engine and seat too if needed. [p. 1-12]

Helicopter tie-down: moor it facing the highest forecast wind, spot it more than one rotor span from other aircraft, and tie down the blades and fuselage per the manufacturer.



Figure 1-13. Example of mooring a helicopter.

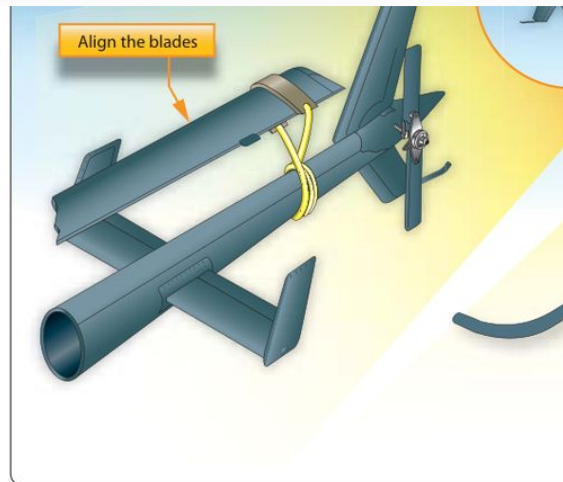


Figure 1-14. Securing helicopter blades and fuselage.

MUST KNOW **Tie-down numbers:** head **into the wind**; manila rope needs about **1 inch of slack** (it shrinks wet); **never tie to a lift strut**; use a **bowline** (anti-slip) knot; helicopters survive **~65 mph** tied down and spot **> one rotor span** from other aircraft; seaplane storm move = **flood the floats**; ski plane = **“dead-man”** buried in the snow.

[pp. 1-10 to 1-12]

PART 5 — Ground Movement of Aircraft

Before starting ANY aircraft engine (applies to all powerplants): [p. 1-13]

1. **Head the aircraft into the prevailing wind** for cooling airflow over the engine.
2. Make sure prop blast / jet exhaust won't cause property damage or injury.
3. If using external electrical power, make sure it can be removed safely and is enough for the whole start.
4. **Station a “fireguard” with a suitable fire extinguisher** — a CO₂ extinguisher of **at least 5-pound capacity**, placed adjacent to the outboard side of the engine, in the pilot's view.
5. For turbine aircraft, keep the area in front of the jet inlet clear of personnel, property, and debris (FOD).
6. Follow the manufacturer's checklists for start and shutdown.

Reciprocating engines — start cautions

- **Hydraulic lock:** before starting a radial engine that sat **more than 30 minutes**, ignition OFF, then **turn the prop 3–4 complete revolutions by hand** to detect a hydraulic lock (liquid pooled in a lower cylinder). A lock shows up as abnormal effort or the prop stopping abruptly. **Never force the prop** — you can bend or break a connecting rod. To clear it,

remove a lower-cylinder spark plug and pull the prop through; never turn it backward (that pushes liquid into the intake). [p. 1-13]

- **Mixture:** general rule — **idle cut-off for fuel injection, full rich for float-type carburetors.** [p. 1-13]
- Throttle to a position giving **1,000–1,200 rpm**; leave carb heat / alternate air in **COLD** to prevent fire damage if it backfires. [p. 1-13]
- **Starter duty cycle:** don't crank continuously for more than **1 minute**; allow a **3- to 5-minute cooling period** between attempts or you can burn out the starter. [p. 1-14]
- After it starts: move mixture to full rich (if started in idle cut-off) and **check oil pressure — if it doesn't show within 30 seconds, shut down.** Warm-up rpm is usually **1,000–1,300 rpm.** [p. 1-14]
- Keep the **cowl flaps OPEN** during ground running (these engines are air-cooled and depend on airflow). [p. 1-14]

Hand-cranking (hand-propping) dangers

If there's no self-starter, you start by turning the prop by hand. The call-and-response is mandatory: [p. 1-14]

1. The propeller-turner calls: **"Fuel on, switch off, throttle closed, brakes on."** The operator checks each item and **repeats the phrase.**
2. Switch/throttle are NOT touched again until the prop-swinger calls **"contact."** The operator repeats "contact," THEN turns the switch on. **Never turn the switch on and then call "contact."**

Hand-propping traps to remember:

- **Always assume the ignition is ON** while touching a prop — a faulty switch can read OFF and still let current flow (magnetos work by short-circuiting the current to stop), so the engine could fire. [p. 1-14]
- Be sure the ground is firm (no grass/mud/grease/gravel). **Swing the blade downward with the palms;** never curl fingers over the edge — a **kickback** can break them or pull you into the blades. Stand close enough to step away as the blade comes down. [p. 1-14]
- **Over-priming** causes fires, scored cylinders/pistons, and hydraulic lock. If flooded: ignition OFF, throttle FULL OPEN, turn it over to clear fuel — if excessive force is needed, STOP. [p. 1-14]

Extinguishing engine fires (induction / exhaust fire response)

A fireguard with a **CO₂** extinguisher stands by during every start, and must know the engine's induction system so he/she can direct CO₂ into the **air intake**. [p. 1-15]

1. **If an induction fire starts during cranking, keep cranking to start the engine and blow out the fire.**
2. If it won't start and the fire continues, discontinue the start; the fireguard then extinguishes it with the available equipment.

Turboprop starting (the 3 steps every turbine start follows)

Every turbine start is **three steps in sequence**: (1) the **starter** spins the compressor to make airflow; (2) at the right speed the **igniters** fire a hot spark; (3) **fuel** is introduced and lights off. As it accelerates to a **self-sustaining speed**, the starter is disengaged. [p. 1-15]

- **Fixed turbine**: prop is connected directly, so it turns during start (extra drag). **Free turbine**: no mechanical link between gas generator and power turbine, so the prop stays feathered and only turns as the gas generator accelerates. [p. 1-15]
- **Starters**: small engines use an electric motor (or starter/generator); big engines use an **air turbine starter** fed by an APU, a ground air cart, or engine cross-bleed. [p. 1-15]
- **While starting, always**: observe the starter duty cycle; confirm enough air pressure / electrical capacity; **do NOT start if turbine inlet (residual) temperature is above the manufacturer's limit**; provide fuel under low pressure to the fuel pump. [p. 1-15]

Turbofan starting

Turbine aircraft need no preflight run-up unless investigating a malfunction. Remove covers, head into wind, clear the run-up area of personnel and loose items, check fuel sumps for water/ice, and inspect the inlet/fan blades for FOD and damage. [pp. 1-16 to 1-17] Typical sequence: power lever to idle → **boost pumps on** → confirm fuel inlet pressure → starter on (engine rotates, check oil pressure) → ignition on → start lever to idle/start (fuel flows) → **light-off = a rise in exhaust gas temperature** → check N₁ (fan) rotation and oil pressure → starter off at proper speed. [p. 1-17]

Unsatisfactory turbine starts (hot start / hung start)

Bad start	What happens	Cause & what to do
Hot start	Engine starts but exhaust gas temperature exceeds limits	Caused by an excessively rich fuel/air mixture — too much fuel or not enough airflow. Shut the fuel off immediately.
False / hung start	Engine starts normally but rpm stays at a low value	Usually insufficient starter power , or the starter cut off before the engine self-accelerated. Shut the engine down.

Bad start	What happens	Cause & what to do
	and won't rise to normal starting rpm	
Fails to start	Doesn't start within the time limit	Lack of fuel, no/insufficient power to the ignition exciter, or wrong fuel mixture. Shut down; in all unsatisfactory starts turn off fuel and ignition , then motor the compressor ~15 sec to clear fuel (or wait 30 sec to drain if you can't motor it).

MUST KNOW **Hot start** = EGT too high, from too-rich mixture → **shut fuel off NOW**.
Hung (false) start = engine lit but rpm **stuck low**, from weak starter → **shut it down**.

[p. 1-17]

Auxiliary Power Units (APUs)

An APU is generally a **small turbine engine** that supplies **compressed air for starting engines, cabin heating/cooling, and electrical power on the ground**. Turn the switch to start (spring-loaded), and it starts automatically while you watch exhaust gas temperature. APUs run at **idle at 100% rpm with no load**, and are normally used to start the main engines. [p. 1-17]

Towing of aircraft (wing walkers & limits)

Towing is hazardous if careless. Large aircraft are moved with a tow tractor ("tug") and a **tow bar**. [pp. 1-19 to 1-20]

- **A qualified person must be in the flight deck to work the brakes** in case the tow bar fails or unhooks. [p. 1-19]
- **Tow bars are built for tension (pulling) only** — not for torsional/twisting loads. [p. 1-19]
- **Wing walkers:** a wing walker is stationed at **each wingtip** to ensure clearance; a **tail walker** is added for sharp turns or backing. [p. 1-20]
- **Never walk or ride between the nosewheel and the towing vehicle**, and don't ride on the aircraft or tug or board/leave them while moving. [p. 1-20]
- **Towing speed must not exceed the walking team members' pace**. Don't start/stop suddenly; brakes are applied only on a team member's command in an emergency. [p. 1-20]
- Charge the brake system before towing; keep **chocks immediately available**; check tires and struts for proper inflation first. [p. 1-20]
- **Tail-wheel aircraft:** tow forward by attaching the bar to the **main landing gear**, and **unlock the tail wheel** first (or you can break the locking mechanism). **Tricycle gear:** tow forward by the **nosewheel axle**. [pp. 1-19 to 1-20]

pushing on the correct areas of the aircraft. Aircraft also be taxied about the flight line but usually only by ce qualified personnel.



Figure 1-19. Example of a tow tractor (tug).

y The attachment of the tow bar varies on different types of
n aircraft. Aircraft equipped with tail wheels are generally
towed forward by attaching the tow bar to the main landing
gear. In most cases, it is permissible to tow the aircraft in



Figure 1-20. Example of a tow bar.

Taxiing aircraft

Only rated pilots and qualified A&P technicians are authorized to start, run up, and taxi aircraft. [p. 1-20] A taxi signalman helps when the pilot's view is blocked; the signalman stands **slightly ahead of and in line with the aircraft's left wingtip** (the nose is on the signalman's left). Rule of thumb: **if you can see the pilot's eyes, the pilot can see your signals.** [p. 1-21] Day signals follow the FAA AIM; the night stop signal is the "emergency stop" (crossed lighted wands forming an "X"). [pp. 1-21 to 1-22]

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Figure 1-23. The taxi signalman.

Tower light	Meaning
Flashing green	Cleared to taxi
Steady red	Stop
Flashing red	Taxi clear of the runway in use
Flashing white	Return to starting point
Alternating red and green	Exercise extreme caution

Standard taxi light signals (this table reproduces the chapter's **Figure 1-22**). [\[p. 1-21, Fig. 1-22\]](#)

Hand signals: the chapter shows the standard FAA hand taxi signals and the helicopter operating signals. Learn these — they show up on the written. At night the signalman uses lighted wands.

[\[pp. 1-21 to 1-22\]](#)

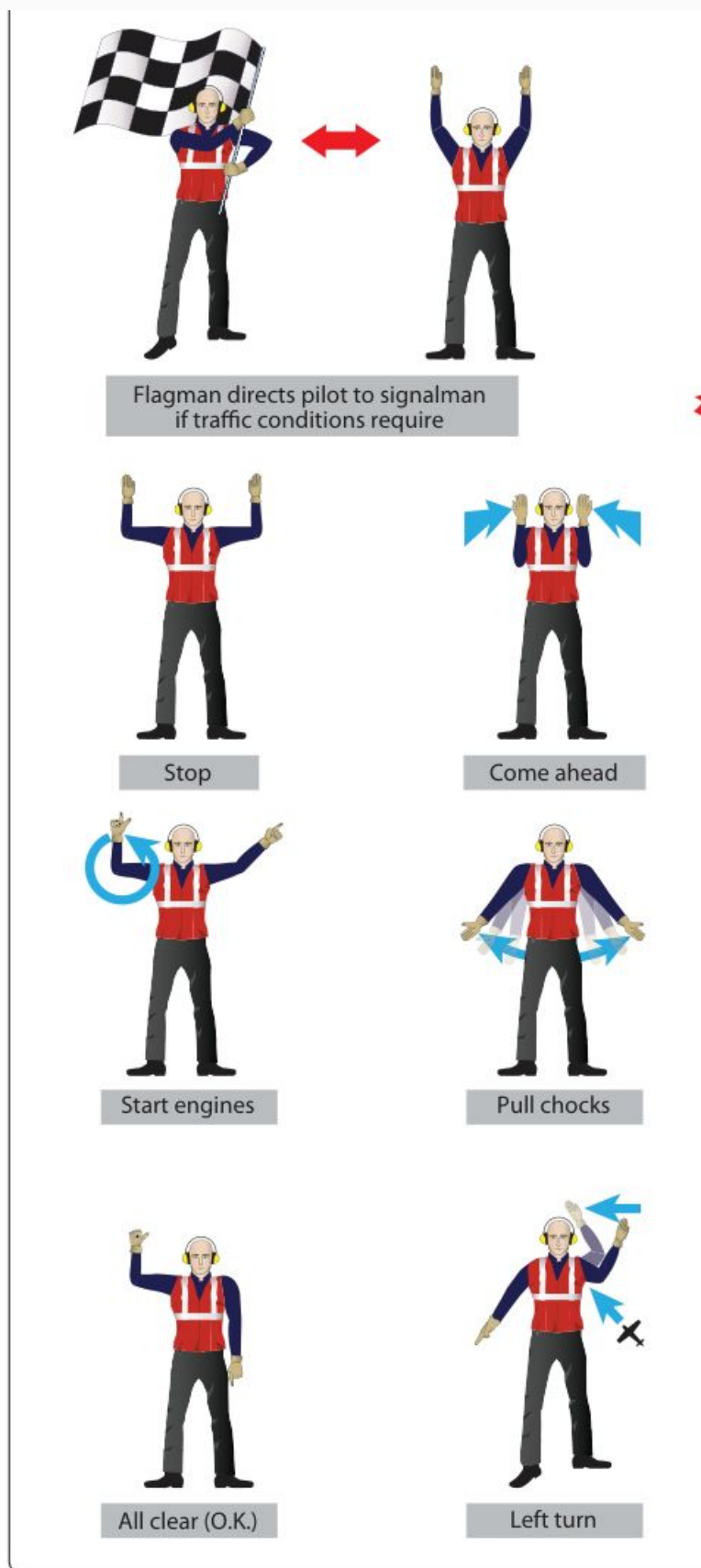


Figure 1-24. Standard FAA hand taxi signals.

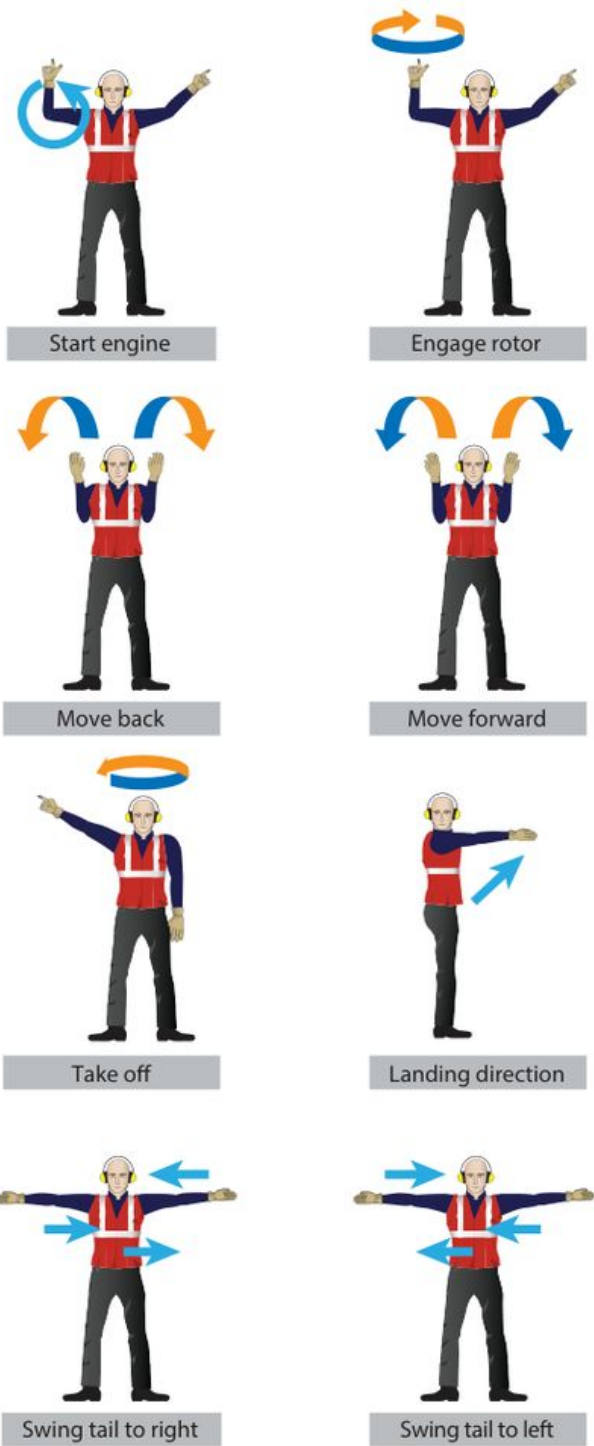


Figure 1-25. Helicopter operating signals.



Figure 1-26. Night operations with lighted wands.

PART 6 — Servicing Aircraft

Before servicing any aircraft fluid, consult the specific maintenance manual for the proper equipment and procedures. [p. 1-22]

Oil and fluids

- Engine oil is checked with a **dipstick or sight gauge**. **Reciprocating engines: check after the engine has been inactive. Turbine engines: check just after shutdown.** [p. 1-22]
- **Dry-sump systems hide oil** that seeped from the tank into the gear case — it won't show on the dipstick until the engine is run/motored. If you service before that oil returns, you'll **overfill. Never overfill the oil tank** — oil foams as it circulates, and the tank's expansion space is left for that foaming. Always use the correct oil for the engine. [p. 1-22]
- Spilled oil/fuel/hydraulic fluid on skin or clothes must be removed ASAP (fire and health). [p. 1-22]
- **Hydraulic reservoir:** use the correct fluid (identify by container or color). Some reservoirs are air-pressurized — bleed the pressure before servicing. If you change filters, ensure system pressure is OFF first; after major fluid loss, **purge/bleed air** and check for leaks. [pp. 1-23 to 1-24]

Servicing with air / nitrogen

When servicing tires or struts with **high-pressure nitrogen**, use caution: **clean the area before connecting the filling hose and do not over-inflate.** [p. 1-24]

Ground Support Equipment (GSE)

- **Electric ground power units (GPUs)** come as **towed, stationary, or self-propelled**. Supply **constant-current/variable-voltage DC for turbine starts** and **constant-voltage DC for reciprocating starts**. Position the unit so it can't collide with the aircraft if its brakes

fail; extend the cable near full length but don't stretch it. **Never move a power unit while service cables are attached or the generator is engaged.** [pp. 1-24 to 1-25]

- **Hydraulic ground power units (“hydraulic mule”)** drain, filter, refill, and test the aircraft hydraulic system. **At 3,000 psi a leak stream can cut like a sharp knife** — inspect lines for cuts/frays and keep them kink-free. [p. 1-25]
- **Ground support air units (air carts)** provide **low-pressure (up to 50 psi), high-volume air** for engine starting and cabin heating/cooling. [p. 1-25]

Oxygen servicing equipment & OXYGEN HAZARDS

- **Two people are required to service gaseous oxygen:** one at the equipment control valves, one watching the aircraft system pressure, in communication. [p. 1-25]
- **Do NOT service oxygen during fueling, defueling, or any work that could be an ignition source.** Service oxygen **outside the hangar.** [p. 1-26]
- Oxygen comes as **gaseous** (steel cylinders) or **liquid (LOX)** (stored/converted in a liquid-oxygen converter). Only **“Aviator’s Breathing Oxygen” (Federal Spec BB-o-925A, Grade A)** goes in aircraft breathing systems — NOT industrial (impurities) or medical (water that freezes). [p. 1-26]
- **LOX causes severe “burns” (frostbite) on skin contact.** [p. 1-26]

MUST KNOW OXYGEN — NO OIL/GREASE RULE: gaseous oxygen is chemically stable and nonflammable BY ITSELF, but it makes other materials **burn faster and hotter**, and it **combines with oil, grease, or bituminous material to form a highly EXPLOSIVE mixture** that's sensitive to impact or compression. So **keep oil and grease completely away from oxygen** and only let qualified people service it. [p. 1-26]

5th-grade picture: Plain air is about 1/5 oxygen. Pure oxygen is like pouring rocket-fuel air on a flame — a tiny spark that would barely burn in normal air becomes a roaring fire. Add a smear of oil or grease and it can **explode just from being squeezed or bumped**. That's why “no oil near oxygen” is an iron rule.

PART 7 — Fuel Servicing

Types of fuel and identification

Two general types: **aviation gasoline (AVGAS)** for reciprocating engines, and **turbine fuel / jet fuel** for turbine engines. AVGAS grades are written as two numbers = the **lean-mixture and rich-mixture octane ratings** (e.g. 80/87 = 80 lean, 87 rich). Grades are also color-coded. [p. 1-26]

Fuel	Color	Used in / notes
AVGAS 80 (80/87)	 Red	Reciprocating engines (lowest lead)
AVGAS 100 (100/130)	 Green	Reciprocating engines
AVGAS 100LL (low lead)	 Blue	Reciprocating engines — the common general-use grade today
AVGAS 115 (115/145)	 Purple	Limited use in large reciprocating-engine aircraft
Jet A / Jet A-1	 Black band (fuel itself clear/straw)	Turbojet/turboshaft. Made from kerosene . (Jet B = kerosene + AVGAS blend.)

The fuel's color must match the color band on the piping and fueling equipment. [p. 1-26, Fig. 1-31]

A large amount of water is drained after fueling an aircraft. Entrained water is ended water droplets. These droplets may not be visible

Color	Grade
Red	80
Green	100
Blue	100LL
Purple	115

Figure 1-31. Aviation gasoline color and grade reference.

MUST KNOW AVGAS color code (memorize both directions): **80 = Red · 100 = Green · 100LL = Blue · 115 = Purple**. Jet fuel = **Black** band on equipment (actual fuel is clear/straw). The two AVGAS numbers = **lean / rich octane ratings**. [p. 1-26]

Exam trap — never mix fuels: Before ever mixing AVGAS and turbine fuel, check the Type Certificate Data Sheet. **Jet fuel in AVGAS** → less power and possible **detonation** (engine damage, loss of life). **AVGAS in jet fuel** → **lead deposits** in the turbine and reduced service life. [p. 1-26]

Contamination control (water, solids, microbes)

Contamination is anything in the fuel that shouldn't be there. The three types: **water, solids, and microbial growth.** The best control is to **prevent it from getting in**; filters, separators, and screens catch most of the rest. [p. 1-26]

- **Water** — two forms: **dissolved (vapor)** and **free water.** Dissolved water isn't a big problem until cold turns it into free water that can form **ice crystals and clog filters.** Free water shows as **water slugs** (concentrations you drain after fueling) or **entrained water** (suspended droplets that make fuel look cloudy and settle out over time). [pp. 1-26 to 1-27]
- **Solids** — insoluble bits: rust, dirt, sand, gasket material, lint, shop-towel fragments. Particles as small as **1/20 the diameter of a human hair** can damage fuel controls. [p. 1-27]
- **Microbiological growth** — micro-organisms living in the **free water in jet fuel**; they grow in warm temperatures with iron oxide and mineral salts present. They form **slime/sludge that fouls filters**, emulsify the fuel, and make **corrosive compounds** that attack the tank. **The best prevention is to keep the fuel dry.** [p. 1-27]

Fueling hazards — volatility & static

- **Volatility** = a liquid's ability to change into a gas at a relatively low temperature. **Liquid fuel doesn't burn — the VAPOR does.** That vapor is what powers the engine and what creates the fire hazard. [p. 1-27]
- **Static electricity** is made by one substance rubbing another — fuel flowing through a line, and especially an aircraft moving through the air in flight, builds a static charge. If that charge isn't dissipated before refueling, it tries to return to ground through the fuel line, and **the spark can ignite fuel vapor.** That's why we bond and ground. [p. 1-27]

Static / bonding & grounding during fueling (the order matters)

When fueling with mobile (truck) equipment, the chapter spells out the sequence: [pp. 1-27 to 1-29]

1. Approach with caution; position the fuel truck so it can **drive away forward** (no backing) if needed. Set the hand brake and chock the wheels.
2. **Ground the aircraft, then ground the truck, then bond them together** with a connecting wire (three separate ground wires, or a "Y" cable).
3. Make sure grounds touch **bare metal or proper grounding points. Do NOT use the engine exhaust or propeller as grounding points.**
4. **Ground the nozzle to the aircraft, THEN open the fuel tank.**
5. Protect the wing and other items from spilled fuel and careless handling.
6. Check the fuel cap for proper installation/security before leaving.
7. **Remove the grounding wires in the REVERSE order** they were attached. (If the aircraft won't fly/move soon, the aircraft ground wire may be left attached.)

MUST KNOW **Bonding & grounding order:** ground the **aircraft** → **then the truck** → **bond them together** → **then ground the nozzle to the aircraft** → **then open the tank**. Remove the wires in **reverse**. **Never** ground to the exhaust or propeller. The reason for all of it: **a static spark can ignite fuel vapor**. [pp. 1-27 to 1-29]

Fueling procedures & the pre-fuel checklist

Two basic methods: small aircraft use the **over-the-wing** method (hose into ports on top of the wing); large aircraft use **single-point (pressure) fueling** — receptacles in the bottom leading edge fill all tanks at once, which is faster, limits contamination, and **reduces the chance of static igniting the fuel**. [p. 1-27]

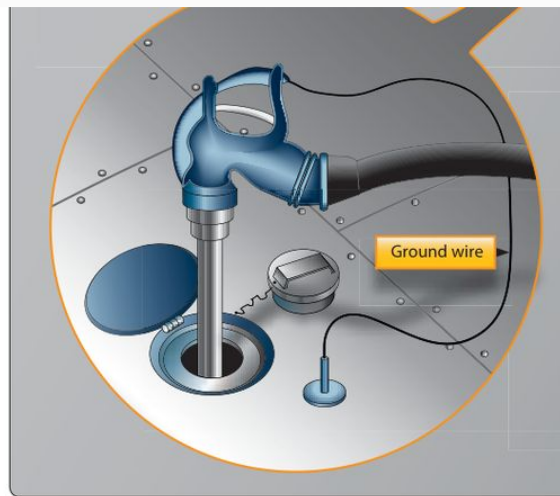


Figure 1-32. Refueling by the over-the-wing method.



Figure 1-33. Single-point refueling station of a large aircraft.

Before fueling, the person fueling must: [pp. 1-27 to 1-29]

- Turn **all aircraft electrical systems and electronic devices (including radar) OFF**.
- Carry **nothing in shirt pockets** (it could fall in the tank) and **no flame-producing devices**.
- Verify the **proper type and grade of fuel — do not mix AVGAS and JET**.
- Ensure **all sumps have been drained**. Wear **eye protection** (gloves/apron help too).
- Don't fuel if nearby aircraft could blow dirt onto the open tank. **Don't fuel with lightning within 5 miles. Don't fuel within 500 feet of operating ground radar.**

MUST KNOW **Fueling distance limits:** no fueling with **lightning within 5 miles**, and no fueling **within 500 feet of operating ground radar**. Turn **all electrical/electronic gear (incl. radar) OFF** and drain the **sumps** first. [pp. 1-27 to 1-29]

Defueling

Procedures differ by aircraft — **always check the maintenance/service manual first**. Defuel either by **gravity** (need a way to collect the fuel) or by **pumping** (don't damage the tanks; the removed fuel **cannot be mixed back with good fuel**). General precautions: [p. 1-29]

- **Ground the aircraft and the defueling equipment.**
- **Turn off all electrical and electronic equipment.**
- Have the **correct type of fire extinguisher available**.
- **Wear eye protection.**

One-breath recap of the whole chapter: Keep the shop clean and respect electricity, compressed gas, and machine tools (eye protection always). On the flight line: hearing protection near loud tools, control **FOD**, and respect prop/intake/exhaust and rotor danger zones (helicopters: in the pilot's view, never from the rear, go around the nose). Fire needs **oxygen + heat + fuel**; classes **A/B/C/D** each get a matching extinguisher (water=A, CO₂=A/B/C, Halon=B/C, dry powder=D); use it with **PASS**. Tie aircraft **into the wind**. Start engines with a **CO₂ fireguard**; watch for hydraulic lock, hand-prop discipline, and turbine **hot/hung starts**. Tow with **wing walkers** and brakes manned. Service fluids per the manual; **no oil/grease near oxygen**; know **AVGAS colors** (80-red, 100-green, 100LL-blue, 115-purple, jet-black) and **bond/ground** before fueling.

Built by amt-tutor from the assigned chapter read in full (FAA-H-8083-30B Ch.1, pp. 1-1 to 1-30). This is a study summary, not a sign-off for real maintenance. Worksheet answers should match your AIM slide wording where it differs; the written-exam wording follows the FAA handbook as cited above on every fact.